SECPI Manuscript — Editorial Flag Archive

Session pause point: End of Methods section (through §2.6), prior to Results & Discussion
Sections reviewed: Title, Abstract, Introduction, Methods (§2.1–2.6)
Sections pending: Results and Discussion, Conclusion

Locked Decisions (carry forward, no further action needed)

- **Title (final):** *A Generalizable Framework for Synergistic and Equitable Cooling Optimization of Philippine Tree Functional Types via Discrete Grid Modeling*
- **SECPI expansion:** Synergistic and Equitable Cooling Performance Index
- **Species (corrected):**
  - Narra = *Pterocarpus indicus*
  - Akleng-parang = ***Albizia lebbeck*** (correcting an earlier, incorrect web-verification of *A. procera* — your team's source data is authoritative here)
- **Study area:** No real-world site. The domain is a **synthetic, non-georeferenced 100 m × 100 m grid**, generated via Cellular Automata. The only link to real-world Philippine conditions is the land-use ratio parameterization (still pending verification — see #9).
- **Theoretical Foundation (§1.1):** Condensed 3-paragraph version stands in the Introduction; full 4-subsection original to be prepared as **Supplementary Material S1** whenever you're ready.
- **Scope/Delimitations/Spatial Framework/Biological Parameters:** Relocated out of the Introduction into Methods (already integrated in the Methods draft above); Introduction now carries only a short scope statement.
- **Figure captions:** Add "Original artwork by the authors" to Figures 1 and 2 (Canva-original).

OPEN FLAGS — Require Your Input

Organized by status. **VERIFICATION** = your team is still checking; parked but tracked. **NEEDS INPUT** = a decision or data point only you can supply. **ROADBLOCK RISK** = flagged as a VERIFICATION item that, if it fails, sends the section back for rework.

Abstract

#	Flag	Status
5	SECPI score range — should be 0–5, pending team reverification	VERIFICATION
9	Land-use ratio source — confirm specific PH city/region zoning statistics used	VERIFICATION
38	Abstract calls the decay model "Gaussian," but the actual function in §2.3.2 is exponential ($e^{-\lambda d/C_D}$), not Gaussian (e^{-d^2/σ^2}). Affects Abstract wording as well as Methods.	VERIFICATION

Introduction

#	Flag	Status
13	Citation currency (WMO/PAGASA figures are 2018–2023) — on hold per your request, revisit with newer data	VERIFICATION (on hold)
14	2024–2025 citations (Yigitcanlar et al. 2024; Scordato & Gulbrandsen 2024; Abujder Ochoa et al. 2025) unverifiable from PDF alone — confirm author/year/venue	VERIFICATION

Methods

#	Flag	Status
19	"Evergreen tree types" claim — *Terminalia catappa* (Talisay) and *Lagerstroemia speciosa* (Banaba) are typically deciduous/semi-deciduous in the literature; check against your source	VERIFICATION
20	AGB estimation error percentages (50% at 10×10m, 10% at 50×50m, 5% at 100×100m) — no in-text citation for the PTM-2/H–D model study; this justifies your entire domain size choice	VERIFICATION
21	"Kunhle et al." (non-submodular canopy interaction citation in §2.1) — name not recognized in the literature, possible misspelling	VERIFICATION
22	NSF (n.d.), EPFL (n.d.) cited as institutions rather than specific works	VERIFICATION
25	CA transition probability equation — same time index $(t+1)$ appears on both sides of $\hat{p}_i^{kl}(t+1) = \gamma[\ldots]p_i^{kl}(t+1)$, and state-label symbols (k, l, j, P) are inconsistent between prose and formula. Needs a check against actual code.	NEEDS INPUT
26	"Expander heuristic" — confirm whether this term is drawn directly from Almeida et al. (2002) or coined by your team for this adaptation	VERIFICATION
27	Equity/vulnerability definition in §2.2.3 closely echoes phrasing already used in Introduction §1.1.3 (Resilience Thinking)	DEFERRED — low priority, address in final consistency pass
28	Coarse grid vs. fine grid described with **identical** dimensions (10×10 cells, 10 m^2/cell) despite being called "lower-resolution" and "higher-resolution" respectively. Also conflicts with Figure 4 (100m×100m domain ÷ 10m cells = 100 m^2/cell, not 10 m^2).	**NEEDS INPUT — ROADBLOCK RISK**
29	"cooling effects of a blanking configuration" — likely typo for "planting configuration"; quick confirm	NEEDS INPUT (minor)
30	DBH-from-height formula ($DBH = (h/h_0)^{1/h_1}$, citing Fischer et al./FORMIND) — typical FORMIND height–diameter allometrics run DBH→height, not height→DBH. Confirm the equation wasn't inverted in transcription.	VERIFICATION
31	Species-specific parameters (h_0, h_1, l_0, l_1) for all six TFTs — needed as a table (Methods or Appendix) for reproducibility	NEEDS INPUT (data readiness)
33	Fine-grid cell size restated a **third** way in §2.4.2: "each cell represents 1 m^2" vs. "10×10 fine evaluation grid (10 m^2 resolution)" — same underlying issue as #28, now with a	**NEEDS INPUT — ROADBLOCK RISK,**

#	Flag	Status
	third conflicting value in the same paragraph	**extends #28**
34	§2.4 states ACO is used to "minimize" SECPI; §2.4.2 states the objective is "Maximize f(x)...= SECPI." A positive SECPI score is explicitly framed as improvement. Likely a one-word slip in §2.4, but confirm which direction is correct in your actual implementation.	NEEDS INPUT (quick confirm)
35	Sigmoidal decay calibration: stated "≈62% of maximum cooling at the crown edge" doesn't follow from the derived $\lambda = 1.897$ (calculation gives ≈38.7% at $d/C_D = 0.5$). Recheck the arithmetic or the definition of "crown edge."	VERIFICATION
36	"Shaamala, 2025" (§2.4 intro) vs. "Shaamala (2024)" (§2.4.1, §2.4.2) — confirm actual publication year(s); may be two distinct sources	VERIFICATION
37	"Chebyshev space ($\mathbb{Z}^2$)" used to describe the tree-placement lattice in §2.4.2, while the cooling decay function elsewhere explicitly uses Euclidean distance (d_{ij}). Confirm intended metric.	VERIFICATION
39	"Statistically significant redirection of resources" (§2.5.2, cross-scenario validation) — test type, replicate count, and significance threshold not yet specified	NEEDS INPUT (data readiness)
41	$CCA_{threshold}$ and steepness parameter k given only as illustrative examples ("e.g., 1.2") — actual fixed values used in the model need to be reported for reproducibility	NEEDS INPUT (data readiness)

Quick-Reference Counts

- **Total flags raised across Abstract/Intro/Methods:** 41
- **Resolved / locked:** 24
- **Still open:** 17 (3 Abstract, 2 Introduction, 12 Methods)
- **Highest-priority open items:** #28 / #33 (grid resolution — same underlying issue, appears 3 times with conflicting values; recommend resolving as one unit against your actual code before Results, since Results will report metrics computed on this grid)

Resuming This Session

When you're ready to continue with Results and Discussion:

1. Send the section pages as before.
2. Flag numbering continues from **42**.
3. Let me know if any of the VERIFICATION items above have resolved, become ROADBLOCKS, or are still pending — I'll fold updates into the tracker as we go.
4. If #28/#33 (grid resolution) gets resolved in the meantime, flag it first — it may affect how Results reports spatial metrics.